

STAT 639 - Homework 1

ISLR Chapter 2: Exercises 1, 2, 8, and 10

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1 Exercise 1

For each case, determine whether a more flexible statistical learning method would generally be expected to perform better or worse than a less flexible method.

1.1 (a) Large number of observations, small number of predictors

A more flexible method should generally perform better.

When n is very large and p is small, there is enough data to estimate a flexible model without suffering too much from high variance. A flexible model can capture more complicated relationships between the predictors and response, so the reduction in bias may outweigh the increase in variance.

1.2 (b) Small number of observations, large number of predictors

A more flexible method should generally perform worse.

With small n and large p , a flexible method can easily fit random noise in the training data. This leads to overfitting and high variance. A less flexible method is generally safer in this setting.

1.3 (c) Highly non-linear relationship between predictors and response

A more flexible method should generally perform better.

If the true relationship is highly non-linear, a flexible model is better able to approximate that relationship. A simple model may have high bias because it cannot capture the true structure.

1.4 (d) Very high variance of the error terms

A more flexible method should generally perform worse.

When the irreducible noise is very large, a highly flexible method may begin fitting the noise rather than the underlying signal. This increases variance and can hurt test-set performance.

2 Exercise 2

For each situation, identify whether the problem is classification or regression, whether the main goal is inference or prediction, and identify n and p .

2.1 (a) CEO salary

The response is CEO salary, and the explanatory variables include company profit, number of employees, and industry.

- **Type:** Regression
- **Goal:** Inference
- n : 500 firms
- p : 3 predictors

This is a regression problem because CEO salary is quantitative. The purpose is to understand which factors are associated with CEO salary, so the primary goal is inference.

2.2 (b) Success of a new product

The response indicates whether a product is successful or unsuccessful. The predictors consist of price, marketing budget, competitors' price, and ten other variables.

- **Type:** Classification
- **Goal:** Prediction
- n : 20 products
- p : 13 predictors

This is classification because the response is qualitative. The goal is to predict whether the new product will succeed.

2.3 (c) Weekly change in the U.S. dollar

The response is the weekly percentage change in the U.S. dollar relative to the euro, using weekly percentage changes in the stock market and the company's market as predictors.

- **Type:** Regression
- **Goal:** Prediction
- n : 52 weeks
- p : 2 predictors

The response is quantitative, so this is regression. Since the goal is to forecast the weekly percentage change, the main goal is prediction.

3 Exercise 8 - College Data

This exercise uses the `College` data set described in ISLR Chapter 2.

3.1 (a) Read the data

The code below first looks for `College.csv` in the current working directory. If it is not present, it downloads the file from the book website.

```
college_file <- "College.csv"

if (!file.exists(college_file)) {
  download.file(
    "https://www.statlearning.com/s/College.csv",
    destfile = college_file,
    mode = "wb"
  )
}

college <- read.csv("College.csv", check.names = FALSE)
```

There are 777 colleges and universities. Before removing the institution-name column, the imported data contain 19 columns.

3.2 (b) Use college names as row names

```
# Inspect first column
head(college[[1]])

## [1] "Abilene Christian University" "Adelphi University"
## [3] "Adrian College"             "Agnes Scott College"
## [5] "Alaska Pacific University"   "Albertson College"
anyDuplicated(college[[1]])

## [1] 0

# Only use it as row names if it is unique
if (anyDuplicated(college[[1]]) == 0) {
  rownames(college) <- college[[1]]
  college <- college[, -1, drop = FALSE]
}else{
  rownames(college) <- college[, 1]
  college <- college[, -1]
}

dim(college)

## [1] 777 18

head(college)

##               Private Apps Accept Enroll Top10perc Top25perc
## Abilene Christian University   Yes 1660 1232 721      23      52
## Adelphi University            Yes 2186 1924 512      16      29
## Adrian College                Yes 1428 1097 336      22      50
## Agnes Scott College           Yes 417 349 137      60      89
## Alaska Pacific University      Yes 193 146 55      16      44
## Albertson College             Yes 587 479 158      38      62
##               F.Undergrad P.Undergrad Outstate Room.Board Books
## Abilene Christian University 2885      537 7440      3300 450
```

## Adelphi University	2683	1227	12280	6450	750	
## Adrian College	1036	99	11250	3750	400	
## Agnes Scott College	510	63	12960	5450	450	
## Alaska Pacific University	249	869	7560	4120	800	
## Albertson College	678	41	13500	3335	500	
##	Personal	PhD	Terminal	S.F.Ratio	perc.alumni	Expend
## Abilene Christian University	2200	70	78	18.1	12	7041
## Adelphi University	1500	29	30	12.2	16	10527
## Adrian College	1165	53	66	12.9	30	8735
## Agnes Scott College	875	92	97	7.7	37	19016
## Alaska Pacific University	1500	76	72	11.9	2	10922
## Albertson College	675	67	73	9.4	11	9727
##	Grad.Rate					
## Abilene Christian University	60					
## Adelphi University	56					
## Adrian College	54					
## Agnes Scott College	59					
## Alaska Pacific University	15					
## Albertson College	55					

After removing the institution-name column, the data contain 777 observations and 18 variables. The college names are now stored as row names rather than as a predictor.